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Technical proposal (Envelope 1)

Tender NGPA/T/2026/021 · Port Community System: Web Platform, Messaging Hub and Integration Services

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/021, we, Qasr Digital Systems L.L.C., submit our technical proposal for the Port Community System. This envelope contains no prices. We confirm that our bid is valid as stated in our financial proposal.

Khalid Al-Rashdi
Managing Director, Qasr Digital Systems L.L.C.

Date: 12 January 2027
Reference: QDS-PCS-2027-004

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1. Company and references

Qasr Digital Systems is a systems integrator registered in the State since 2011, with 140 staff in Al-Noor and the capital. We implement the PortLink PCS product under a regional licence from its developer and operate the product's regional support centre.

Reference	Scope	Year
Port of Ras Hadd PCS	Portal, EDIFACT hub, TOS and customs integration, 2,900 users	2023
Free Zone Logistics Portal	Truck appointments for 3 gates, Arabic/English app	2024
Inland Depot Authority	Messaging hub upgrade, 180,000 messages/day	2025

2. Solution and architecture

The PCS runs entirely in the Authority's two data centres: an active site and a standby site with database replication. All components (portal, messaging hub, integration adapters, monitoring, backup) are installed on Authority infrastructure. Licences are perpetual and activated with an offline licence file; updates are delivered as signed packages through the Authority's change process. The system has no runtime dependency on any external service.

The messaging hub converts UN/EDIFACT (BAPLIE, COPARN, CODECO, CUSCAR, CUSRES), XML and JSON/REST with guaranteed delivery, retry queues and end-to-end message tracking. Integration adapters are included for both terminal operating systems, the Customs Risk Management System, the Maritime Single Window and the Authority's ERP.

Sizing: 6 application servers and a 3-node database cluster per site. Our load test on this configuration (report attached as Appendix A) reached 5,000 registered users, 800 concurrent sessions and 300,000 messages per day with a 95th-percentile processing time of 3.2 seconds.

3. Security (CITS-2026)

- Internal users sign in through the Authority's Active Directory over LDAPS; roles map to job functions. External users use multi-factor authentication (authenticator app or SMS one-time code); MFA is also enforced for every administrative account.
- Local accounts: minimum 12 characters, lock-out after 5 failed attempts, session lock after 15 minutes of inactivity.
- Audit logs of all user and administrator actions are hash-chained, kept for 5 years and forwarded in real time to the Authority's SIEM in CEF format.
- AES-256 encryption at rest; TLS 1.3 in transit with certificates from the Authority's PKI.
- Critical vulnerabilities fixed within 14 days and high within 30 days, during the whole support period.
- An independent penetration test by a CREST-accredited firm before go-live; critical and high findings closed before acceptance.
- Remote support only through the Authority's VPN, approved per session and recorded.
- Daily encrypted backups replicated to the standby site; RPO 4 hours, RTO 4 hours.
- Our administrators will complete the Authority's security vetting and sign its confidentiality undertaking.

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- A CycloneDX SBOM is delivered with every release; architecture, data-flow and hardening documents before the Site Acceptance Test.

4. Implementation plan and support

Milestone	Month
Design sign-off	2
Portal and messaging hub in production	7
TOS, customs, Maritime Single Window and ERP integrations live	9
Vehicle booking system and haulier app (Android, iOS) live	11
Data migration of 3 years of vessel calls	8

Training: 30 Authority administrators and helpdesk staff (2 courses), and onboarding sessions for 160 community companies in Arabic and English.

Support: 24/7 service desk in Al-Noor. Severity 1: 30-minute response and 4-hour restoration. Availability commitment 99.8 % per calendar month for the portal and the messaging hub, measured by the Authority's monitoring system, with service credits.

Dashboards for container dwell time, gate waiting times and release lead times are part of the standard product. All REST APIs are published in OpenAPI 3 with a sandbox environment for community developers. The whole user interface, including the haulier app, is available in Arabic (right-to-left) and English.

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5. Compliance with RFP sections 3 and 4

Ref	Requirement	Response	Offered
3.1	Hosting in the Authority's data centres in the State; no public cloud dependency	Comply	Both Authority data centres, offline licence
3.2	Availability 99.8 % per calendar month (portal and messaging hub)	Comply	99.8 % monthly, service credits
3.3	4,500 users, 600 concurrent, 250,000 messages/day, p95 < 5 s	Comply	Tested 5,000 / 800 / 300,000 per day, p95 3.2 s
3.4	MFA for all external users; Active Directory for internal users	Comply	MFA (app or SMS) external; AD over LDAPS internal
3.5	Full Arabic and English UI incl. right-to-left	Comply	Arabic RTL and English
3.6	Open REST APIs (OpenAPI 3) with sandbox	Comply	OpenAPI 3, sandbox
3.7	Dashboards: dwell time, gate waiting, release lead times	Comply	Standard dashboards
3.8	Haulier mobile app (Android and iOS)	Comply	Android and iOS
3.9	Compliance with CITS-2026	Comply	See section 3
4.1	Go-live portal, hub, customs/TOS integrations ≤ 10 months; VBS ≤ 12 months	Comply	Integrations month 9, VBS month 11
4.2	Training 30 staff; onboarding ≥ 150 companies	Comply	30 staff, 160 companies
4.3	24/7 severity-1: 1 h response, 4 h restoration; 3 years support priced per year	Comply	30 min / 4 h, priced per year

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